THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x = 3$ is a zero of $x^2 - 13x + 30$, the other zero is: 1

(a) 27 **(b)** -10 **(c)** 16 **(d)** 10

2. If α, β are zeroes of $x^2 - 7x + 10$, then $\alpha^3 + \beta^3$ is: 1

(a) 203 **(b)** 553 **(c)** 343 **(d)** 133

3. If α, β are zeroes of $x^2 - 4x - 12$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) -48 **(b)** -8 **(c)** 144 **(d)** 16

4. If α, β are zeroes of $x^2 - 14x + 33$, then $|\alpha - \beta|$ is: 1

(a) 8 **(b)** 9 **(c)** 14 **(d)** 16

5. If α, β are zeroes of $x^2 - 7x + 10$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is: 1

(a) $\frac{7}{10}$ **(b)** $\frac{29}{10}$ **(c)** $\frac{49}{10}$ **(d)** 29

6. For the cubic with zeroes $1, -2, 6$, $\alpha\beta + \beta\gamma + \gamma\alpha$ is: 1

(a) -12 **(b)** -8 **(c)** 5 **(d)** 8

7. If the zeroes of $x^2 - 8x + k$ differ by 2, then k is: 1

(a) 15 **(b)** 14 **(c)** 17 **(d)** 30

8. If $x = 4$ is a zero of $x^2 - 12x + 32$, the other zero is: 1

(a) 8 **(b)** 16 **(c)** 28 **(d)** -8

9. **Assertion (A):** The polynomial $x^2 - x - 6$ has zeros 3 and -2. 1

Reason (R): A quadratic polynomial with sum of zeros S and product of zeros P is $k\,(x^2 - Sx + P)$, $k \neq 0$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The graph of $y = x^2 - 4x + 4$ touches the x-axis at exactly one point. 1
Reason (R): Every quadratic polynomial has two distinct real zeros.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If one zero of $x^2 - 12x + k$ is 2 times the other, find k. 2

12. If α, β are zeroes of $x^2 - 6x + 8$, form a quadratic polynomial whose zeroes are α^2 and β^2. 2

13. If the sum of the zeroes of $x^2 - 13x + k$ is equal to their product, find k. 2

14. If one zero of $x^2 - 15x + k$ is 4 times the other, find k. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find a quadratic polynomial whose zeroes are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, where α, β are zeroes of $x^2 - 7x + 10$. 3

16. The zeroes of $x^2 - 20x + c$ are α and β with $\alpha - \beta = 6$. Find c. 3

17. If α, β, γ are the zeroes of $x^3 - 3x^2 - 6x + 8$, find $\alpha^2 + \beta^2 + \gamma^2$. 3

OR

Find a quadratic polynomial whose zeroes are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, where α, β are zeroes of $x^2 - 7x + 6$. 3

18. The zeroes of $x^2 - 18x + c$ are α and β with $\alpha - \beta = 4$. Find c. 3

OR

If α, β, γ are the zeroes of $x^3 - 9x^2 + 24x - 20$, find $\alpha^2 + \beta^2 + \gamma^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Three students sketch graphs of polynomials. Arjun's parabola opens upwards and just touches the x-axis at $(2, 0)$. Bela's parabola cuts the x-axis at $(-1, 0)$ and $(3, 0)$ and cuts the y-axis at $(0, -6)$. Chirag's graph is the straight line $y = 2x - 8$. 5
(i) Write the zeros of Bela's polynomial.
(ii) How many zeros does Arjun's polynomial have?
(iii) Find the zero of Chirag's polynomial.
(iv) Find Bela's quadratic polynomial.

20. The zeroes of $x^3 - 18x^2 + 92x - 120$ are in arithmetic progression. Find them. 5

OR

If α, β are the zeroes of $x^2 - 3x - 40$, form a quadratic polynomial whose zeroes are $\alpha + \beta$ and $\alpha\beta$. 5

END OF QUESTION PAPER — Total: 40 Marks